

STATISTICS

PART 4: Normal distribution, Chi-squared distribution, t distribution, F distribution

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Normal Distribution

Statistics

2

A **normal distribution** is a probability density function. Many phenomena in real life (such as the weight of students, and IQ scores of people) can be represented by a normal distribution. A random variable X has a normal distribution if its probability density function is given by

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty < x < \infty \quad -\infty < \mu < \infty, 0 < \sigma < \infty$$

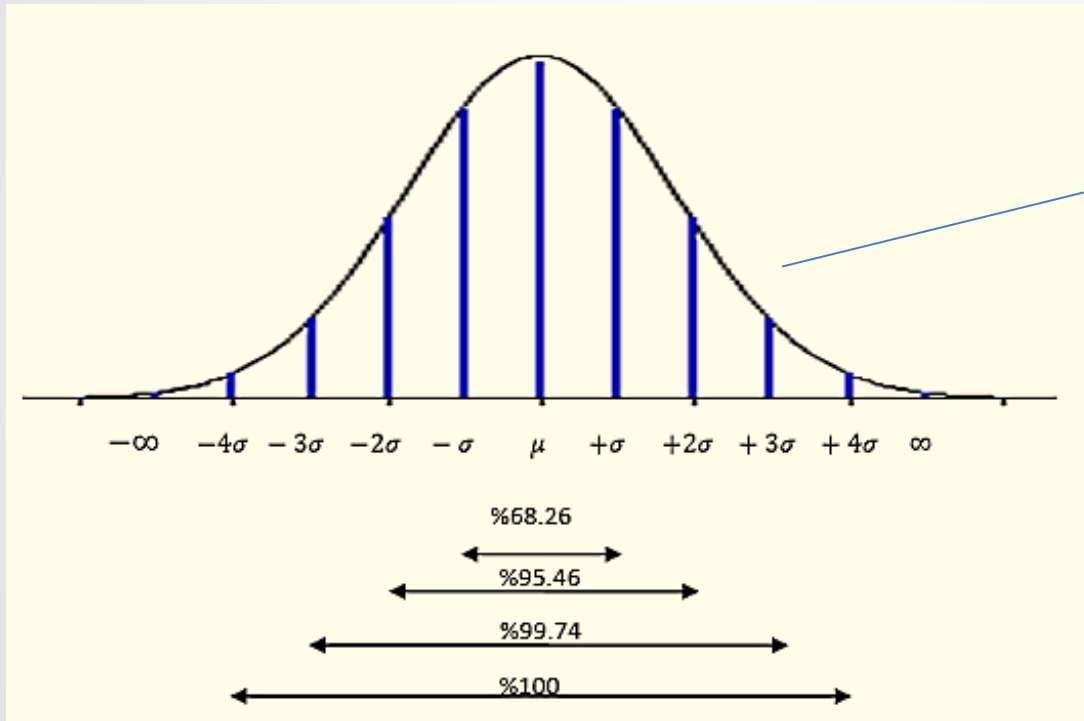
where μ is the mean and σ is the standard deviation of X . To express that X has a normal distribution with mean μ and standard deviation, we use:

$$X \sim N(\mu, \sigma^2)$$

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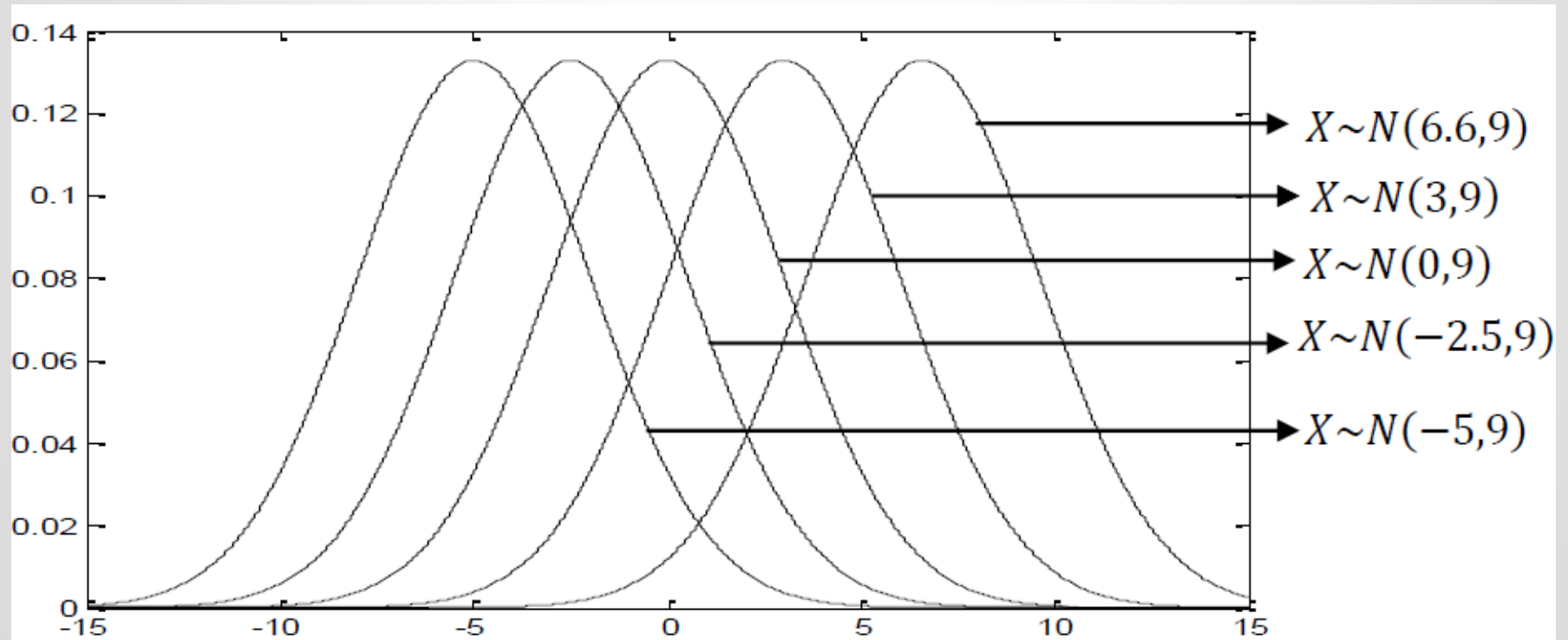
3



A Curve Bell

Normal Distribution

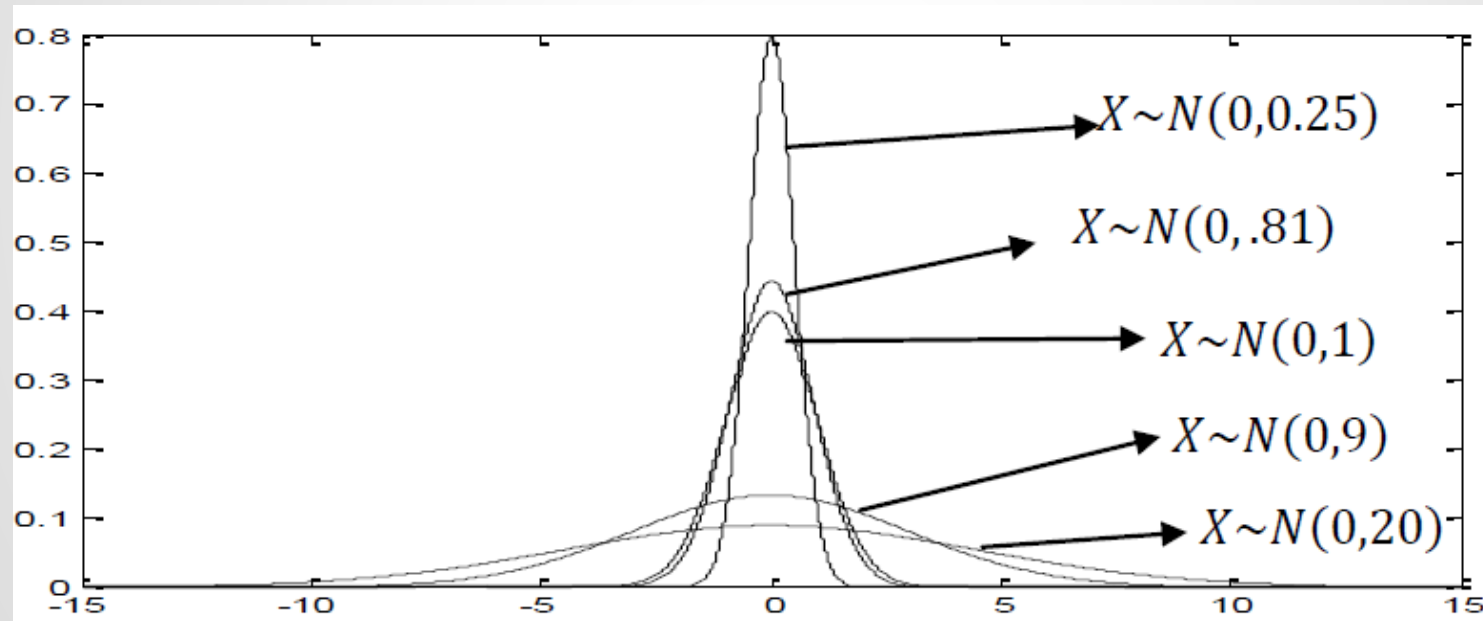
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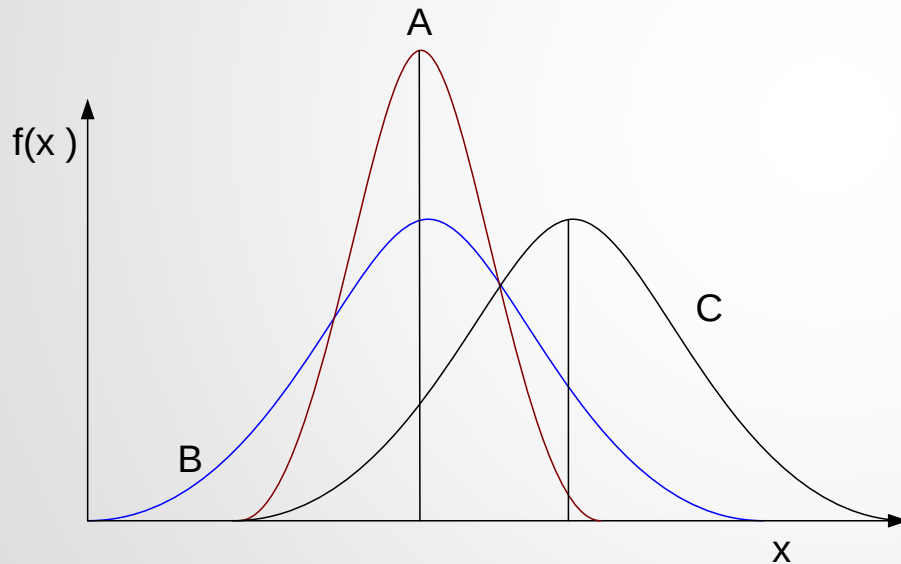
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Normal Distribution

Statistics

6



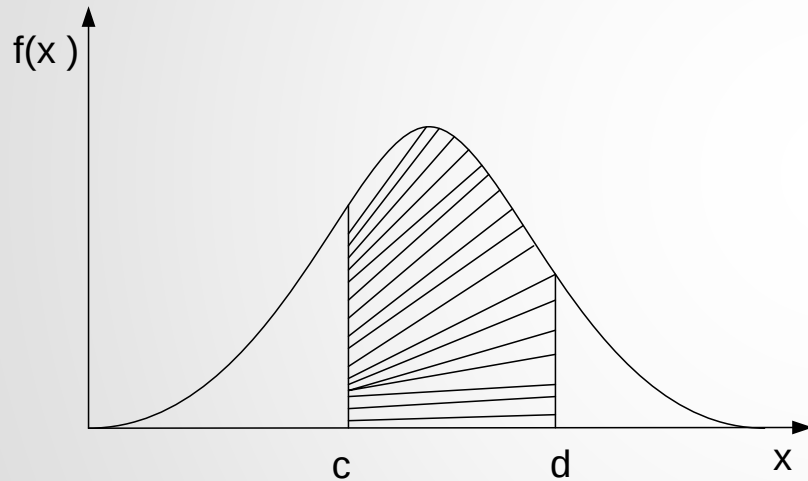
$$\mu_A = \mu_B < \mu_C$$

$$\sigma_A^2 < \sigma_B^2 = \sigma_C^2$$

Normal Distribution

Statistics

7



$$P(c \leq X \leq d) = \int_c^d f(x; \mu, \sigma) dx$$

Normal Distribution

Statistics

Theorem: If $X_1, X_2, \dots, X_n$ are independent random variables distributed normally such as :

$$X_1 \sim N(\mu_1, \sigma_1^2), \quad X_2 \sim N(\mu_2, \sigma_2^2), \quad \dots, \quad X_n \sim N(\mu_n, \sigma_n^2)$$

then the linear combination:

$$Y = c_1 X_1 + c_2 X_2 + \dots + c_n X_n = \sum_{i=1}^n c_i X_i$$

follows the normal distribution:

$$Y \sim N \left(\sum_{i=1}^n c_i \mu_i, \sum_{i=1}^n c_i^2 \sigma_i^2 \right)$$

Normal Distribution

Statistics

Standard Normal Distribution

A **normal distribution** can be transformed into the **standard normal distribution** by using the standard variable:

$$Z = \frac{X - \mu}{\sigma}$$

In this case;

$$Z = \frac{X - \mu}{\sigma}$$

$$E(Z) = E\left(\frac{X - \mu}{\sigma}\right) = \frac{E(X) - \mu}{\sigma} = \frac{\mu - \mu}{\sigma} = 0,$$

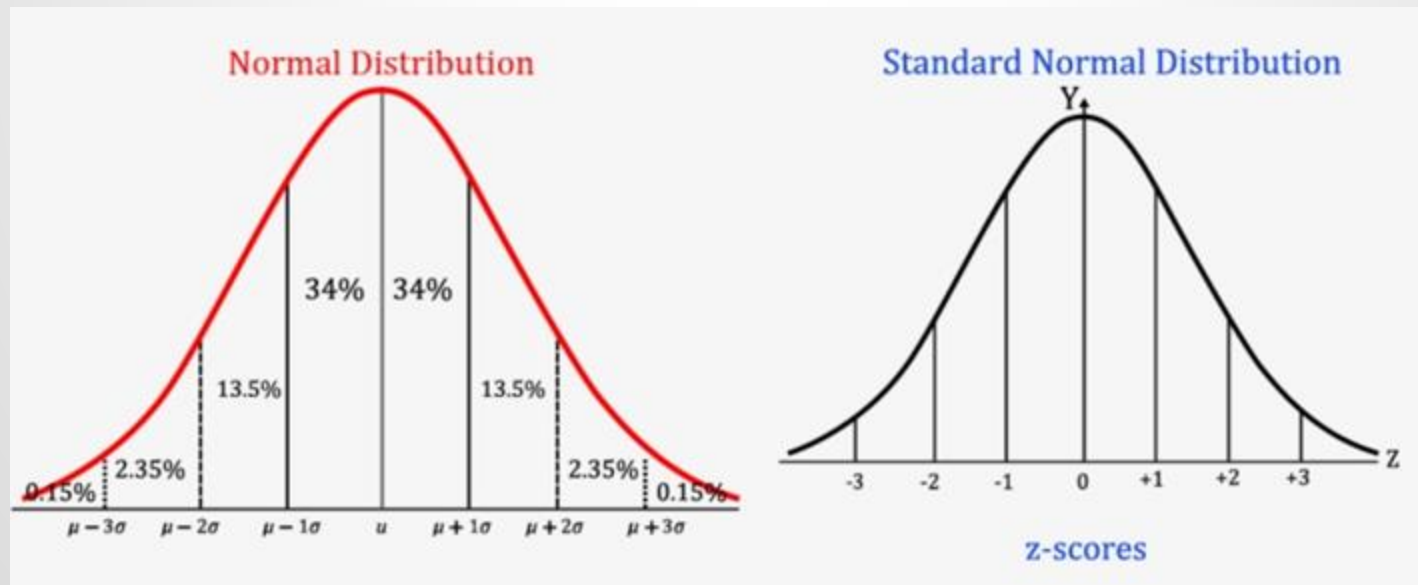
$$\text{Var}(Z) = \text{Var}\left(\frac{X - \mu}{\sigma}\right) = \frac{\text{Var}(X)}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1$$

The mean and standard deviation of the standard normal distribution is $\mu = 0$ and $\sigma = 1$, respectively.

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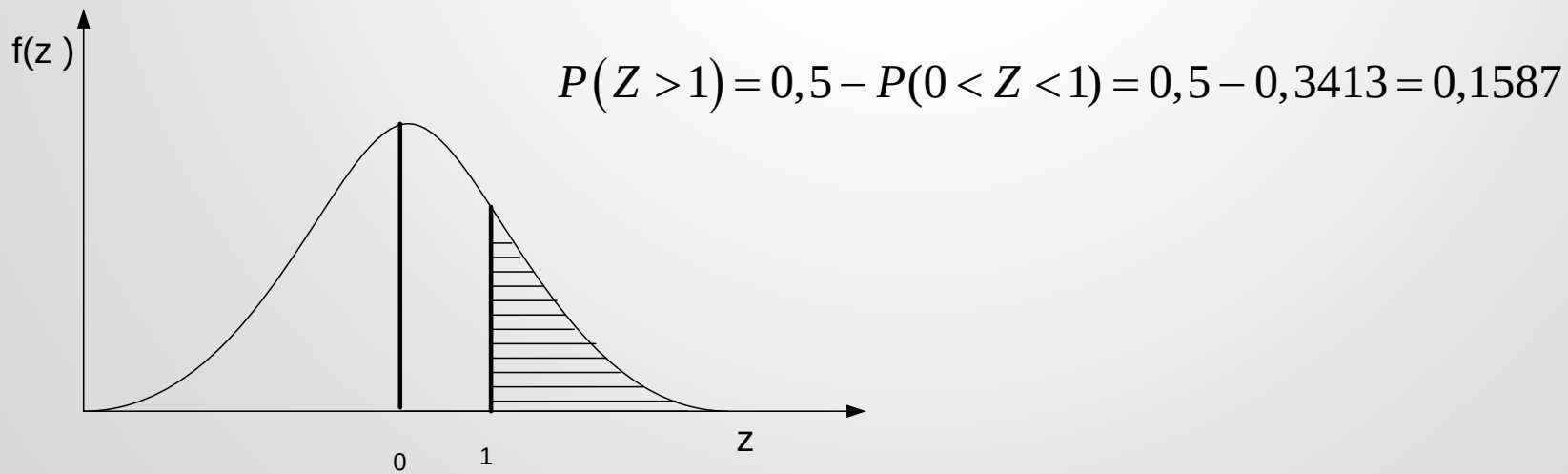
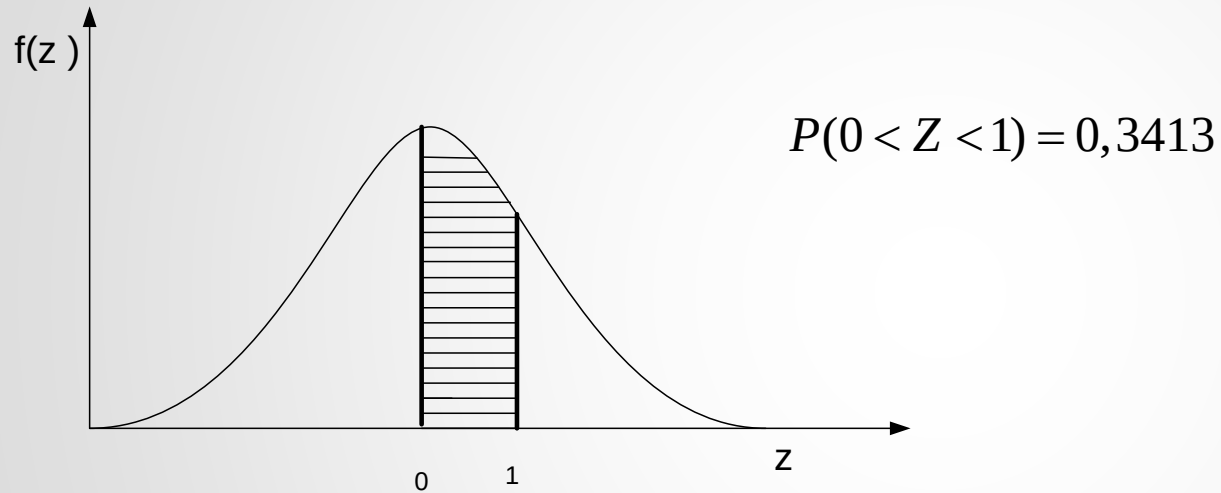
- $X \sim N(\mu, \sigma^2)$
- $Z \sim N(0, 1)$



Normal Distribution

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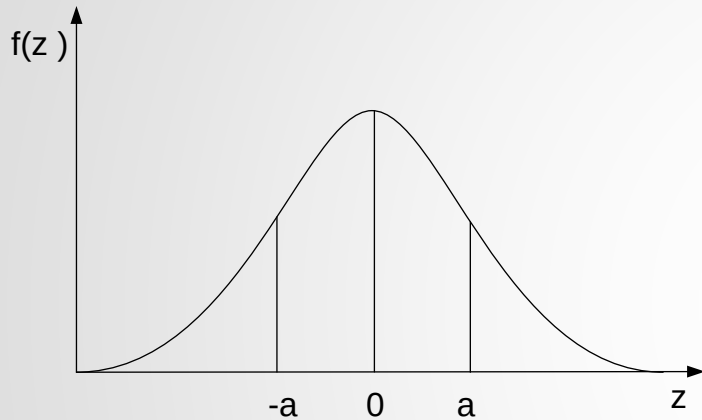
11



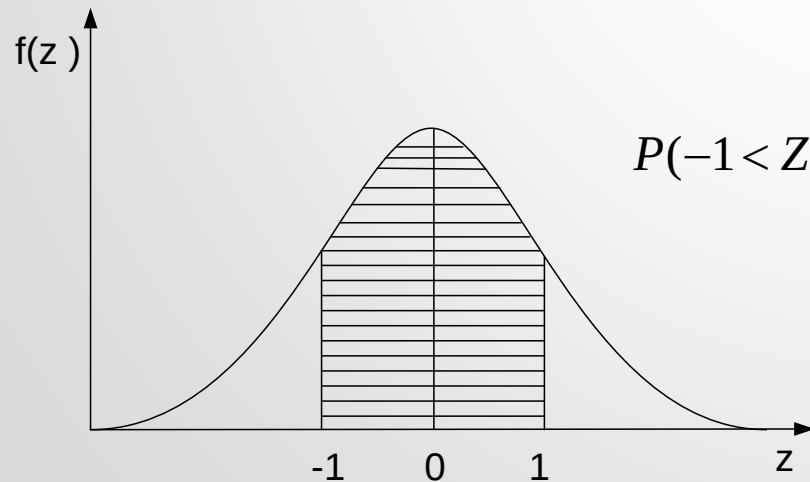
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Statistics

12



$$P(0 < Z < a) = P(-a < Z < 0)$$

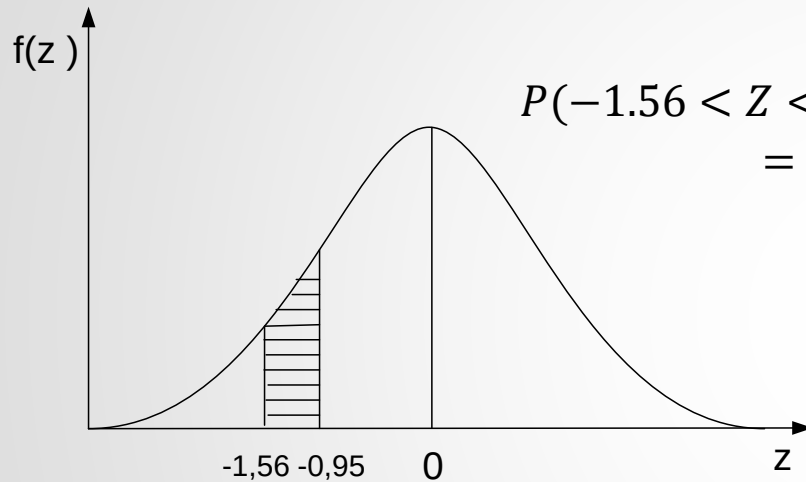


$$\begin{aligned} P(-1 < Z < 1) &= P(-1 < Z < 0) + P(0 < Z < 1) \\ &= 2P(0 < z < 1) = 2(0,3413) = 0,6826 \end{aligned}$$

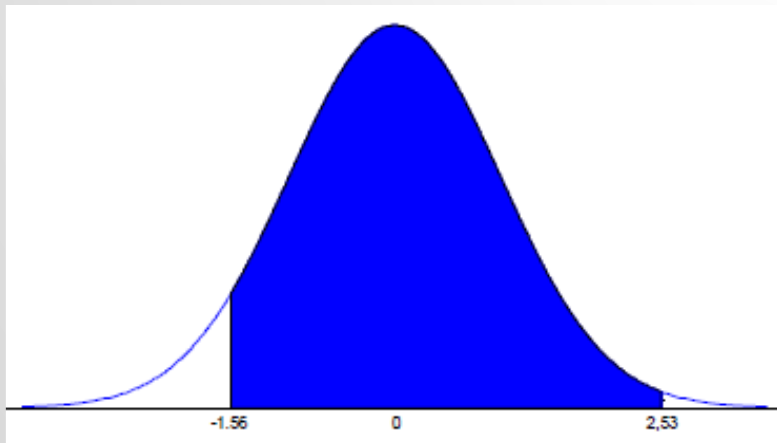
Normal Distribution

Statistics

13



$$\begin{aligned} P(-1.56 < Z < -0.95) &= P(-1.56 < Z < 0) - P(-0.95 < Z < 0) \\ &= 0.4406 - 0.3289 = 0.1117 \end{aligned}$$



$$\begin{aligned} P(-1.56 < Z < 2.53) &= P(-1.56 < Z < 0) + P(0 < Z < 2.53) \\ &= 0.4406 + 0.4942 = 0.9348 \end{aligned}$$

Normal Distribution

Statistics

14

Example 1: If X has a normal distribution with the mean $\mu = 3$ and variance $\sigma^2 = 4$, find $P(3 < X < 5)$?

Ans: 0.3413

Normal Distribution

Statistics

15

Example 2: The weight of the students in a class has a normal distribution with the mean $\mu = 68.5$ and standard deviation $\sigma = 2.3$ kg.

- a) What is the probability that a randomly chosen student in this class weighs more than 72 kg?
- b) What percentage of students' weight is between 70 kg and 72 kg?

Ans: a) 0.0643 b) 19.35%

Normal Distribution

Statistics

16

Example 3: The tea boxes with a capacity of 1 kg are filled by machines, and their weight has a normal distribution with the mean $\mu = 1.03$ kg and the standard deviation $\sigma = 0.02$ kg. What is the probability that a random tea box weighs of

- a) less than 1 kg?
- b) more than 1.06 kg?

Ans: a) 0.0668 b) 0.0668

Normal Distribution

Statistics

17

Example 4: Assume that the life of a certain type of battery produced in a factory is normally distributed. If 33% of the batteries last less than 45 hours and 10% of the batteries last more than 65 hours, find the mean and the standard deviation of the distribution.

Ans: a) $\mu \cong 50.12$ b) $\sigma \cong 11.63$

Normal Distribution

Statistics

18

Example 5: An exam has a normal distribution with a mean of 75 and a standard deviation of 8. The teacher gives A to the students whose grades are 90 or more. If 12 students take A, find the number of students attending the exam.

Ans: a) 400

Other Distributions

Statistics

19

Distributions Derived from Normal Distribution

The following distributions are derived from normal distribution:

➤ Chi-squared distribution

➤ $X \sim \chi_v^2 \implies X$ has a chi-squared distribution with degree of freedom v .

➤ t distribution (student t distribution)

$X \sim t_v \implies X$ has a t distribution with degree of freedom v .

➤ F distribution

$X \sim F_{v_1, v_2} \implies X$ has a F distribution with degree of freedoms v_1 and v_2 .

Chi-Squared Distribution

Statistics

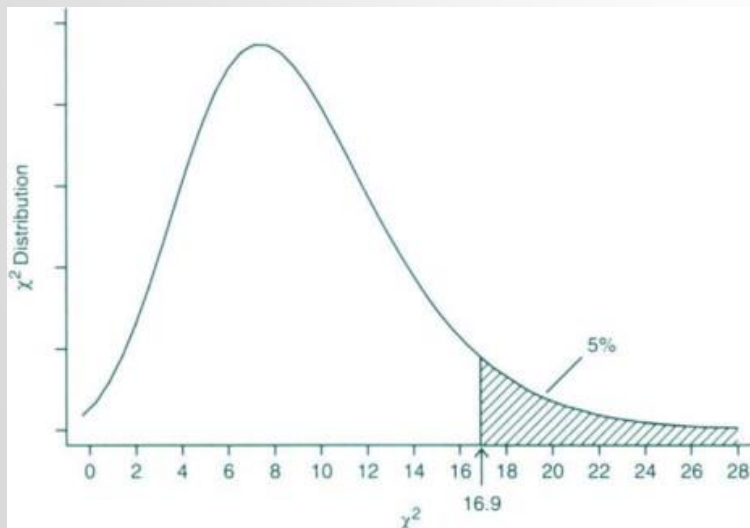
Chi-Squared Distribution

The probability density function of χ^2 (chi-squared) distribution is given by

$$f(x; v) = \frac{x^{v/2-1} e^{-x/2}}{\Gamma(v/2) 2^{v/2}}, 0 \leq x < \infty$$

where positive integer v is called degree of freedom. The mean and variance of the random variable χ^2 is:

$$E(x) = v, \quad \text{Var}(x) = 2v$$



χ^2 distribution is not symmetric and is denoted by χ_v^2 for the degree of freedom v .

Chi-Squared Distribution

Statistics

Theorem: If $Z_1, Z_2, \dots, Z_n$ are independent random variables that follow standard normal distribution, then

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2$$

Theorem: If $X_1, X_2, \dots, X_n$ are independent random variables such that

$$X_1 \sim \chi_{\nu_1}^2, \quad X_2 \sim \chi_{\nu_2}^2, \quad \dots, \quad X_n \sim \chi_{\nu_n}^2$$

then

$$X_1 + X_2 + \dots + X_n = Y \sim \chi_{\nu_1 + \nu_2 + \dots + \nu_n}^2$$

Chi-Squared Distribution

Statistics

For large value of degree of freedom ν , chi-squared distribution approximates normal distribution.

$$\nu \rightarrow \infty \Rightarrow \chi^2_{\nu} \sim N(\nu, 2\nu)$$

Example 6:

- a) $P(2.204 < \chi^2_6 < 10.645) = ?$
- b) If $P(\chi^2_5 < X) = 0.95$, then determine X .

Ans: a) 0.8 b) $X = 11.0705$

Student t Distribution

Statistics

(Student) t Distribution

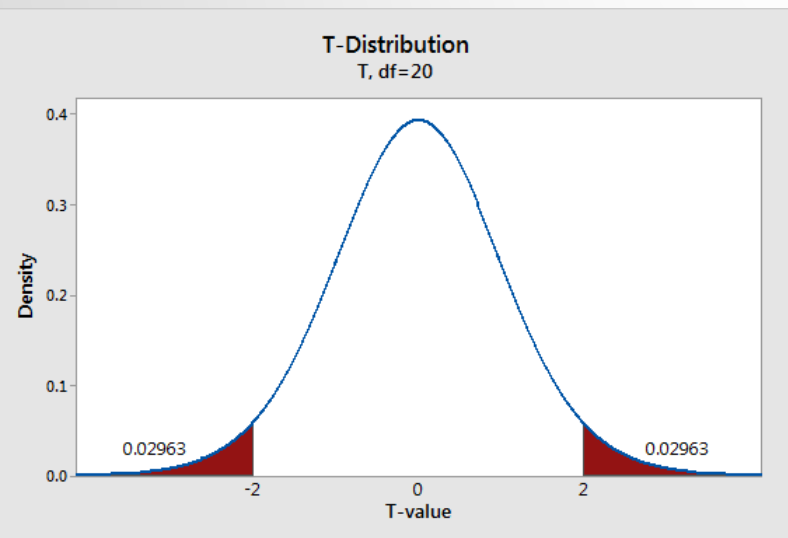
The probability density function of t distribution is given by

$$f(t; v) = \frac{\Gamma\left(\frac{v+1}{2}\right)}{\sqrt{v\pi}\Gamma(v/2)} \left(1 + \frac{t^2}{v}\right)^{-\left(\frac{v+1}{2}\right)}, -\infty \leq t < \infty$$

where positive integer v is called degree of freedom. The mean and variance of the random variable t is:

$$E(t) = 0, \quad \text{Var}(t) = \frac{v}{v-2}$$

t distribution is symmetric and is denoted by t_v for the degree of freedom v .



t Distribution

Statistics

24

Theorem: Let the random variable Z has the standard normal distribution and the random variable Y has a chi-squared distribution with degree of freedom ν . If Z and Y are independent, then

$$t_{\nu} \sim \frac{Z}{\sqrt{Y/\nu}}$$

For large value of degree of freedom ν , t distribution approximates normal distribution.

$$\nu \rightarrow \infty \quad \Rightarrow \quad t_{\nu} \sim N(0,1)$$

t Distribution

Statistics

25

Example 7:

- a) $P(-0.2596 < t_{11}) = ?$
- b) If $P(t_{12} < X) = 0.90$, then determine X .

Ans: a) 0.4 b) $X = 1.3634$

F Distribution

Statistics

F Distribution

The probability density function of *F* distribution is given by

$$f(F; v_1, v_2) = \frac{\Gamma\left(\frac{v_1 + v_2}{2}\right)}{\Gamma\left(\frac{v_1}{2}\right) \Gamma\left(\frac{v_2}{2}\right)} \left(\frac{v_1}{v_2}\right)^{\left(\frac{v_1}{2}\right)} F^{\left(\frac{v_1}{2}-1\right)} \left(1 + \frac{v_1}{v_2} F\right)^{-\left(\frac{v_1+v_2}{2}\right)}, \quad F > 0$$

where positive integer v_1 and v_2 are called degree of freedoms. The mean and variance of the random variable F is:

$$E(F) = \frac{v_2}{v_2 - 2}, \quad \text{Var}(F) = \frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)}.$$

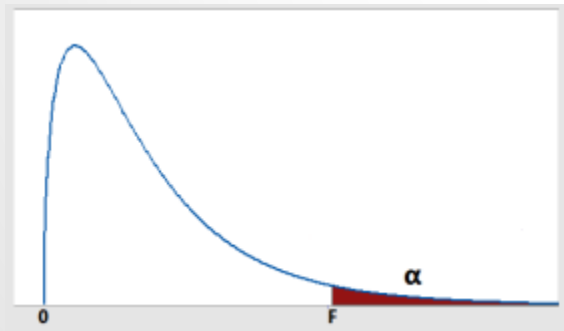
F distribution denoted by F_{v_1, v_2} for the degree of freedom v_1 and v_2 .

F Distribution

Statistics

Theorem: Let the random variable $U \sim \chi_{v_1}^2$ and $V \sim \chi_{v_2}^2$. If they are independent, then

$$F_{v_1, v_2} \sim \frac{U/v_1}{V/v_2}.$$



If $P(X > F_{v_1, v_2}) = \alpha$, then let $F \rightarrow F_{\alpha, v_1, v_2}$

F Distribution

Statistics

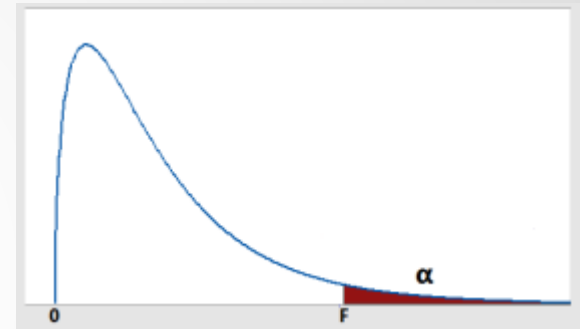
28

Note that;

$$F_{1-\alpha, v_1, v_2} = \frac{1}{F_{\alpha, v_2, v_1}}$$

Example 8:

- a) $P(F_{5,3} < 28.2) = ?$
- b) If $P(F_{6,4} > X) = 0.05$, then determine X .
- c) If $P(F_{6,4} > X) = 0.95$, then determine X .



Ans: a) 0.99 b) $X = 6.16$ c) $X = 1/4.53$